CHAPTER 20

Carboxylic Acids and Nitriles



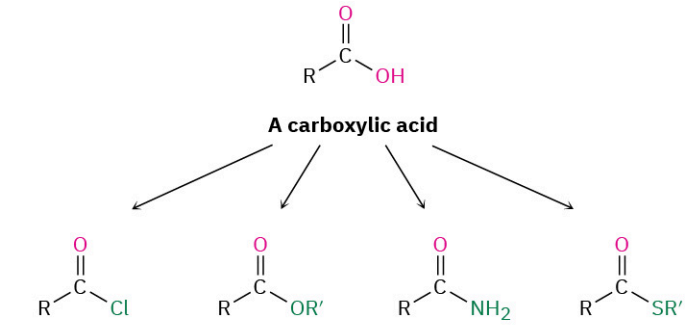
FIGURE 20.1 **The burning sensation produced by touching or eating chili peppers is due to capsaicin, a carboxylic acid derivative called an amide.** (credit: modification of "Chillis" by Lucas Cobb/Flickr, CC BY 2.0)

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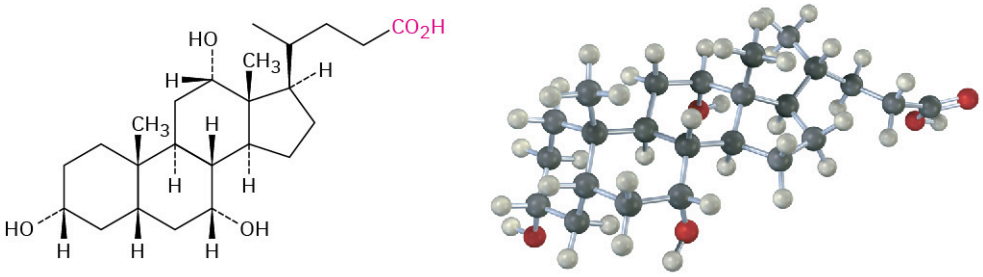
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WHY THIS CHAPTER? Carboxylic acids are present in many industrial processes and most biological pathways and are the starting materials from which other acyl derivatives are made. Thus, an understanding of their properties and reactions is fundamental to understanding organic chemistry. We'll look both at acids and at their close relatives, nitriles (RC≡N), in this chapter and at carboxylic acid derivatives in the next chapter.

Carboxylic acids, RCO_2H, occupy a central place among carbonyl compounds. Not only are they valuable in themselves, they also serve as starting materials for preparing numerous carboxylic acid derivatives such as acid chlorides, esters, amides, and thioesters. In addition, carboxylic acids are present in the majority of biological pathways.

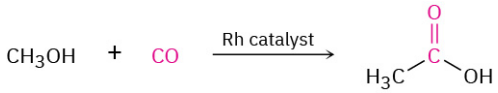


A great many carboxylic acids are found in nature: acetic acid, CH_3CO_2H, is the chief organic component of vinegar; butanoic acid, $CH_3CH_2CH_2CO_2H$, is responsible for the rancid odor of sour butter; and hexanoic acid (caproic acid), $CH_3(CH_2)_4CO_2H$, is responsible for the unmistakable aroma of goats and dirty gym socks (the name comes from the Latin *caper,* meaning "goat"). Other examples are cholic acid, a major component of human bile, and long-chain aliphatic acids such as palmitic acid, $CH_3(CH_2)_{14}CO_2H$, a biological precursor of fats and vegetable oils.



Cholic acid

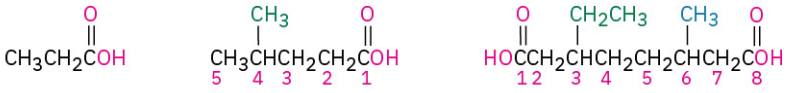
Approximately 20 million tons of acetic acid is produced worldwide each year for a variety of purposes, including preparation of the vinyl acetate polymer used in paints and adhesives. About 20% of the acetic acid synthesized industrially is obtained by oxidation of acetaldehyde. Much of the remaining 80% is prepared by the rhodium-catalyzed reaction of methanol with carbon monoxide.



20.1 Naming Carboxylic Acids and Nitriles

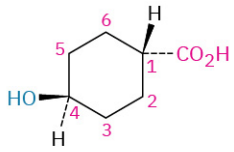
Carboxylic Acids, RCO_2H

Simple carboxylic acids derived from open-chain alkanes are systematically named by replacing the terminal *–e* of the corresponding alkane name with *-oic acid.* The $–CO_2H$ carbon atom is numbered C1.

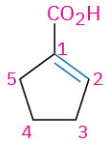


Propanoic acid **4-Methylpentanoic acid** **3-Ethyl-6-methyloctanedioic acid**

Compounds that have a $–CO_2H$ group bonded to a ring are named using the suffix *-carboxylic acid.* The CO_2H carbon is attached to C1 in this system and is not itself numbered. As a substituent, the CO_2H group is called a **carboxyl group**.



***trans*-4-Hydroxycyclohexanecarboxylic acid**



1-Cyclopentenecarboxylic acid

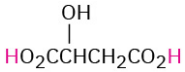
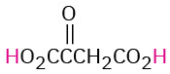
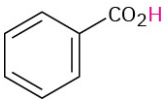
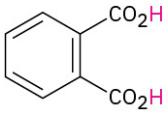
Because many carboxylic acids were among the first organic compounds to be isolated and purified, quite a few common names exist (TABLE 20.1). Biological chemists, in particular, make frequent use of these names, so you may find yourself referring back to this list on occasion. We'll use systematic names in this book, with a few exceptions such as formic (methanoic) acid and acetic (ethanoic) acid, whose names are accepted by IUPAC and are so well known that it makes little sense to refer to them any other way.

Also listed in TABLE 20.1 are the names of acyl groups $\left(R-\overset{\overset{O}{\|}}{C}-\right)$ derived from the parent acids. Except for the eight entries at the top of TABLE 20.1, whose names have a *-yl* ending, all other acyl groups are named using an *-oyl* ending.

TABLE 20.1 Common Names of Some Carboxylic Acids and Acyl Groups

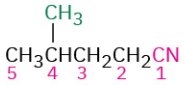
Structure	Name	Acyl group
HCO_2H	Formic	Formyl
CH_3CO_2H	Acetic	Acetyl
$CH_3CH_2CO_2H$	Propionic	Propionyl
$CH_3CH_2CH_2CO_2H$	Butyric	Butyryl
HO_2CCO_2H	Oxalic	Oxalyl
$HO_2CCH_2CO_2H$	Malonic	Malonyl
$HO_2CCH_2CH_2CO_2H$	Succinic	Succinyl
$HO_2CCH_2CH_2CH_2CO_2H$	Glutaric	Glutaryl
$HO_2CCH_2CH_2CH_2CH_2CO_2H$	Adipic	Adipoyl
$H_2C{=}CHCO_2H$	Acrylic	Acryloyl
$HO_2CCH{=}CHCO_2H$	Maleic (cis) Fumaric (trans)	Maleoyl Fumaroyl
$HOCH_2CO_2H$	Glycolic	Glycoloyl
$CH_3\underset{\overset{\|}{OH}}{C}HCO_2H$	Lactic	Lactoyl
$CH_3\overset{\overset{O}{\|}}{C}CO_2H$	Pyruvic	Pyruvoyl
$HOCH_2\underset{\overset{\|}{OH}}{C}HCO_2H$	Glyceric	Glyceroyl

TABLE 20.1 Common Names of Some Carboxylic Acids and Acyl Groups

Structure	Name	Acyl group
	Malic	Maloyl
	Oxaloacetic	Oxaloacetyl
	Benzoic	Benzoyl
	Phthalic	Phthaloyl

Nitriles, RC≡N

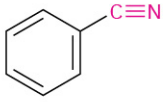
Compounds containing the –C≡N functional group are called **nitriles** and can undergo some chemistry similar to that of carboxylic acids. Simple open-chain nitriles are named by adding *-nitrile* as a suffix to the alkane name, with the nitrile carbon numbered C1.



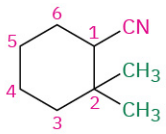
4-Methylpentanenitrile

Nitriles can also be named as derivatives of carboxylic acids by replacing the *-ic acid* or *-oic acid* ending with *-onitrile,* or by replacing the *-carboxylic acid* ending with *-carbonitrile.* The nitrile carbon atom is attached to C1 but is not itself numbered.

Acetonitrile (from acetic acid)

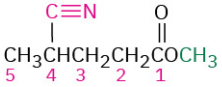


Benzonitrile (from benzoic acid)



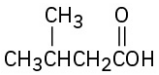
2,2-Dimethylcyclohexanecarbonitrile (from 2,2-dimethylcyclohexanecarboxylic acid)

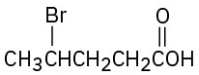
If another carboxylic acid derivative is present in the same molecule, the prefix *cyano-* is used for the C≡N group.

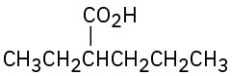


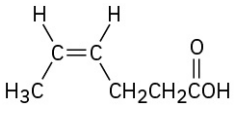
Methyl 4-cyanopentanoate

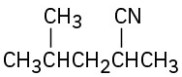
PROBLEM 20-1 Give IUPAC names for the following compounds:

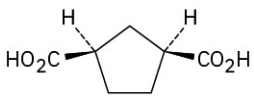
(a) 

(b) 

(c) 

(d) 

(e) 

(f) 

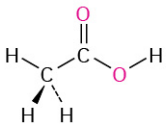
PROBLEM Draw structures corresponding to the following IUPAC names:

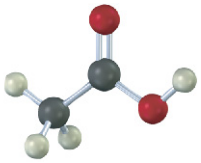
20-2 **(a)** 2,3-Dimethylhexanoic acid **(b)** 4-Methylpentanoic acid
(c) *trans*-1,2-Cyclobutanedicarboxylic acid **(d)** *o*-Hydroxybenzoic acid
(e) (9*Z*,12*Z*)-9,12-Octadecadienoic acid **(f)** 2-Pentenenitrile

20.2 Structure and Properties of Carboxylic Acids

Carboxylic acids are similar in some respects to both ketones and alcohols. Like ketones, the carboxyl carbon is sp^2-hybridized, and carboxylic acid groups are therefore planar with C–C=O and O=C–O bond angles of approximately 120° (TABLE 20.2).

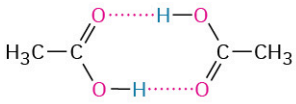
TABLE 20.2 Physical Parameters for Acetic Acid



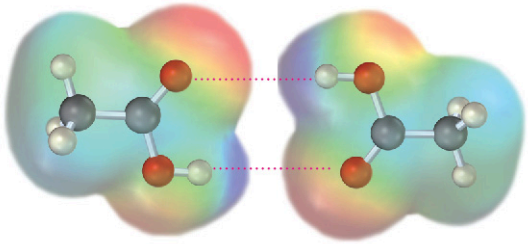


Bond angle	(degrees)	Bond length	(pm)
C–C=O	119	C–C	152
C–C–OH	119	C=O	125
O=C–OH	122	C–OH	131

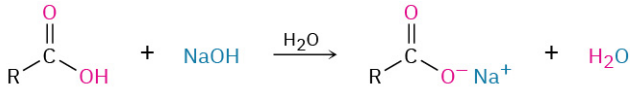
Like alcohols, carboxylic acids are strongly associated because of hydrogen-bonding. Most carboxylic acids exist as cyclic dimers held together by two hydrogen bonds. This strong hydrogen-bonding has a noticeable effect on boiling points, making carboxylic acids boil far less easily than their corresponding alcohols. Acetic acid, for instance, has a boiling point of 117.9 °C, versus 78.3 °C for ethanol, even though both compounds have two carbons.



Acetic acid dimer



The most obvious property of carboxylic acids is implied by their name: carboxylic acids are *acidic*. They therefore react with bases such as NaOH and $NaHCO_3$ to give metal carboxylate salts, $RCO_2^-\ M^+$. Carboxylic acids with more than six carbons are only slightly soluble in water, but the alkali metal salts of carboxylic acids are often highly water-soluble. In fact, it's often possible to purify an acid by extracting its salt into aqueous base, then reacidifying and extracting the pure acid back into an organic solvent.



A carboxylic acid (water-insoluble) **A carboxylic acid salt (water-soluble)**

Like other Brønsted–Lowry acids discussed in Section 2.7, carboxylic acids dissociate slightly in dilute aqueous solution to give H_3O^+ and the corresponding carboxylate anions, RCO_2^-. The extent of dissociation is given by an acidity constant, K_a.



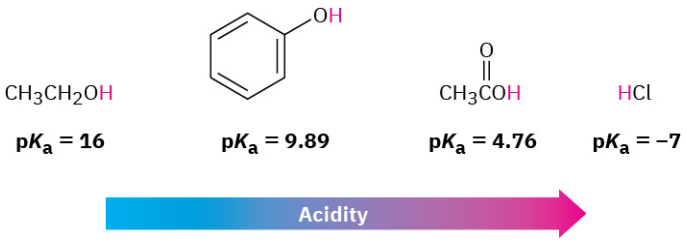
$$K_a = \frac{[RCO_2^-][H_3O^+]}{[RCO_2H]} \quad \text{and} \quad pK_a = -\log K_a$$

A list of K_a values for various carboxylic acids is given in TABLE 20.3. For most, K_a is approximately 10^{-4} to 10^{-5}. Acetic acid, for instance, has $K_a = 1.75 \times 10^{-5}$ at 25 °C, which corresponds to a pK_a of 4.76. In practical terms, a K_a value near 10^{-5} means that only about 0.1% of the molecules in a 0.1 M solution are dissociated, as opposed to the 100% dissociation found with strong mineral acids like HCl.

TABLE 20.3 Acidity of Some Carboxylic Acids

Structure	K_a	pK_a	
CF_3CO_2H	0.59	0.23	
HCO_2H	1.77×10^{-4}	3.75	Stronger acid
$HOCH_2CO_2H$	1.5×10^{-4}	3.84	
$C_6H_5CO_2H$	6.46×10^{-5}	4.19	
$H_2C{=}CHCO_2H$	5.6×10^{-5}	4.25	
CH_3CO_2H	1.75×10^{-5}	4.76	
$CH_3CH_2CO_2H$	1.34×10^{-5}	4.87	Weaker acid
CH_3CH_2OH (ethanol)	(1×10^{-16})	(16)	

Although much weaker than mineral acids, carboxylic acids are nevertheless much stronger acids than alcohols and phenols. The K_a of ethanol, for example, is approximately 10^{-16}, making it a weaker acid than acetic acid by a factor of 10^{11}.



Why are carboxylic acids so much more acidic than alcohols, even though both contain –OH groups? An alcohol dissociates to give an alkoxide ion, in which the negative charge is localized on a single electronegative atom. A carboxylic acid, however, gives a carboxylate ion, in which the negative charge is delocalized over two equivalent oxygen atoms (FIGURE 20.2). In resonance terms (Section 2.4), a carboxylate ion is a stabilized resonance hybrid of two equivalent structures. Since a carboxylate ion is more stable than an alkoxide ion, it is lower in energy and more favored in the dissociation equilibrium.

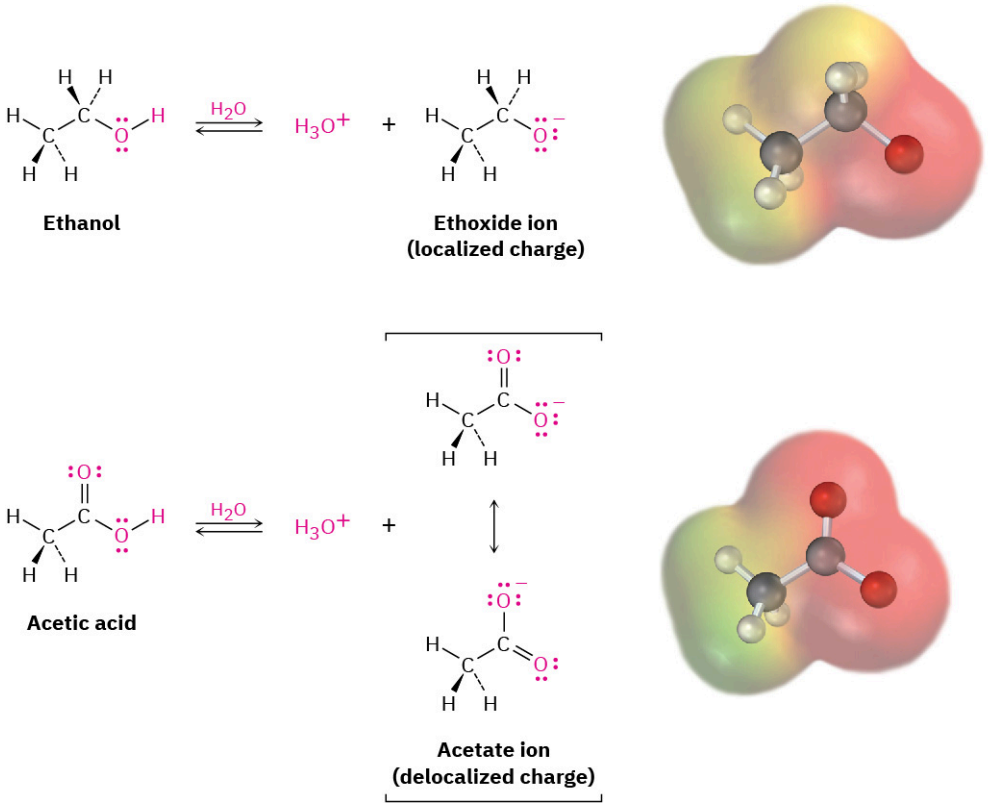
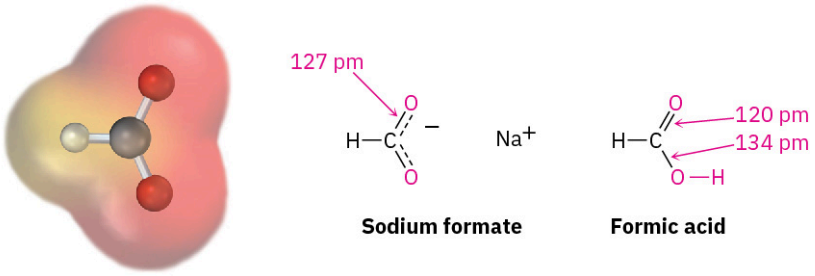


FIGURE 20.2 **An alkoxide ion has its charge localized on one oxygen atom and is less stable, while a carboxylate ion has the charge spread equally over both oxygens and is therefore more stable.**

Experimental evidence for the equivalence of the two carboxylate oxygens comes from X-ray crystallographic studies on sodium formate. Both carbon–oxygen bonds are 127 pm in length, midway between the C=O double bond (120 pm) and the C–O single bond (134 pm) of formic acid. An electrostatic potential map of the formate ion also shows how the negative charge (red) is spread equally over both oxygens.



PROBLEM 20-3 Assume you have a mixture of naphthalene and benzoic acid that you want to separate. How might you take advantage of the acidity of one component in the mixture to effect a separation?

PROBLEM 20-4 The K_a for dichloroacetic acid is 3.32×10^{-2}. Approximately what percentage of the acid is dissociated in a 0.10 M aqueous solution?

20.3 Biological Acids and the Henderson–Hasselbalch Equation

In acidic solution, at low pH, a carboxylic acid is completely undissociated and exists entirely as RCOH. In basic solution, at high pH, a carboxylic acid is completely dissociated and exists entirely as RCO_2^-. Inside living cells, however, the pH is neither acidic nor basic but is instead buffered to a nearly neutral pH of 3.5–4.5 in humans, a value often referred to as *physiological pH*. In what form, then, do carboxylic acids exist inside cells? The question is an important one for understanding the acid catalysts so often found in biological reactions.

If the pK_a value of a given acid and the pH of the medium are known, the percentages of dissociated and

undissociated forms can be calculated using the **Henderson–Hasselbalch equation**.

For any acid HA, we have

$$pK_a = -\log \frac{[H_3O^+][A^-]}{[HA]} = -\log[H_3O^+] - \log \frac{[A^-]}{[HA]}$$
$$= pH - \log \frac{[A^-]}{[HA]}$$

which can be rearranged to give

$$pH = pK_a + \log \frac{[A^-]}{[HA]} \qquad \text{Henderson-Hasselbalch equation}$$

so

$$\log \frac{[A^-]}{[HA]} = pH - pK_a$$

This equation says that the logarithm of the concentration of dissociated acid $[A^-]$ divided by the concentration of undissociated acid [HA] is equal to the pH of the solution minus the pK_a of the acid. Thus, if we know both the pH of the solution and the pK_a of the acid, we can calculate the ratio of $[A^-]$ to [HA]. Furthermore, when pH = pK_a, then HA and A^- are present in equal amounts because log 1 = 0.

As an example of how to use the Henderson–Hasselbalch equation, let's find out what species are present in a 0.0010 M solution of acetic acid at pH = 7.3. According to Table 20-3, the pK_a of acetic acid is 4.76. From the Henderson–Hasselbalch equation, we have

$$\log \frac{[A^-]}{HA} = pH - pK_a = 7.3 - 4.76 = 2.54$$
$$\frac{[A^-]}{HA} = \text{antilog } (2.54) = 3.5 \times 10^2 \quad \text{so} \quad [A^-] = (3.5 \times 10^2)\,[HA]$$

In addition, we know that

$$[A^-] + [HA] = 0.0010\ M$$

Solving the two simultaneous equations gives $[A^-]$ = 0.0010 M and [HA] = 3×10^{-6} M. In other words, at a physiological pH of 7.3, essentially 100% of acetic acid molecules in a 0.0010 M solution are dissociated to the acetate ion.

What is true for acetic acid is also true for other carboxylic acids: At the physiological pH that occurs inside cells, carboxylic acids are almost entirely dissociated. To reflect this fact, we always refer to cellular carboxylic acids by the name of their anion—acetate, lactate, citrate, and so forth, rather than acetic acid, lactic acid, and citric acid.

PROBLEM 20-5 Calculate the percentages of dissociated and undissociated forms present in the following solutions:

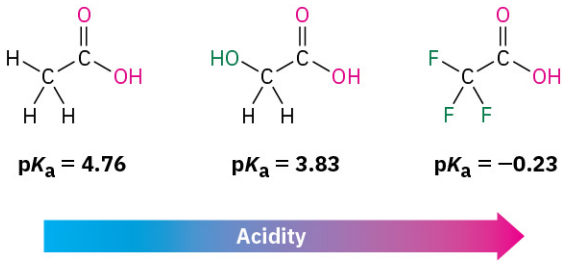
(a) 0.0010 M glycolic acid ($HOCH_2CO_2H$; pK_a = 3.83) at pH = 4.50

(b) 0.0020 M propanoic acid (pK_a = 4.87) at pH = 5.30

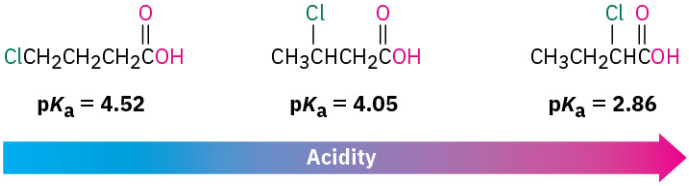
20.4 Substituent Effects on Acidity

The listing of pK_a values shown previously in TABLE 20.3 indicates that there are substantial differences in acidity from one carboxylic acid to another. For example, trifluoroacetic acid (K_a = 0.59) is 33,000 times as strong as acetic acid ($K_a = 1.75 \times 10^{-5}$). How can we account for such differences?

Because the dissociation of a carboxylic acid is an equilibrium process, any factor that stabilizes the carboxylate anion relative to undissociated carboxylic acid will drive the equilibrium toward increased dissociation and result in increased acidity. For instance, three electron-withdrawing fluorine atoms delocalize the negative charge in the trifluoroacetate anion, thereby stabilizing the ion and increasing the acidity of CF_3CO_2H. In the same way, glycolic acid ($HOCH_2CO_2H$; pK_a = 3.83) is stronger than acetic acid because of the electron-withdrawing effect of the electronegative oxygen atom.

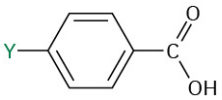


Because inductive effects operate through σ bonds and are dependent on distance, the effect of halogen substitution decreases as the substituent moves farther from the carboxyl. Thus, 2-chlorobutanoic acid has pK_a = 2.86, 3-chlorobutanoic acid has pK_a = 4.05, and 4-chlorobutanoic acid has pK_a = 4.52, similar to that of butanoic acid itself.



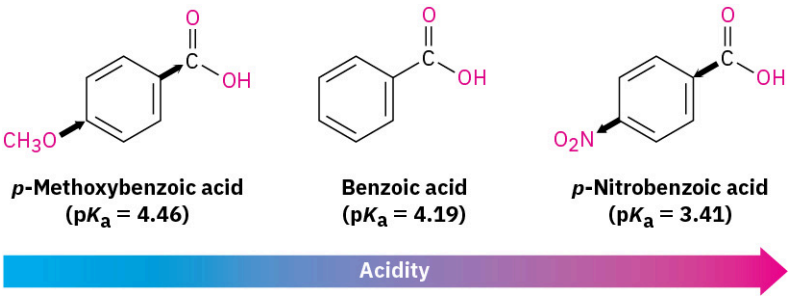
Substituent effects on acidity are also found in substituted benzoic acids. We said during the discussion of electrophilic aromatic substitution in **Section 16.4** that substituents on the aromatic ring strongly affect reactivity. Aromatic rings with electron-donating groups are activated toward further electrophilic substitution, and aromatic rings with electron-withdrawing groups are deactivated. Exactly the same effects can be observed on the acidity of substituted benzoic acids (**TABLE 20.4**).

TABLE 20.4 Substituent Effects on the Acidity of *p*-Substituted Benzoic Acids

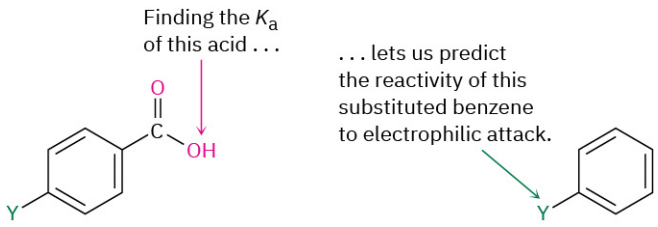


	Y	$K_a \times 10^{-5}$	pK_a	
Stronger acid	$-NO_2$	39	3.41	Deactivating groups
	−CN	28	3.55	
	−CHO	18	3.75	
	−Br	11	3.96	
	−Cl	10	4.0	
	−H	6.46	4.19	
Weaker acid	$-CH_3$	4.3	4.34	Activating groups
	$-OCH_3$	3.5	4.46	
	−OH	3.3	4.48	

As **TABLE 20.4** shows, an electron-donating (activating) group such as methoxy decreases acidity by destabilizing the carboxylate anion, and an electron-withdrawing (deactivating) group such as nitro increases acidity by stabilizing the carboxylate anion.



Because it's much easier to measure the acidity of a substituted benzoic acid than it is to determine the relative reactivity of an aromatic ring toward electrophilic substitution, the correlation between the two effects is useful for predicting reactivity. If we want to know the effect of a certain substituent on electrophilic reactivity, we can simply find the acidity of the corresponding benzoic acid. Worked Example 20.1 gives an illustration.



WORKED EXAMPLE 20.1

Predicting the Effect of a Substituent on the Reactivity of an Aromatic Ring toward Electrophilic Substitution

The pK_a of *p*-(trifluoromethyl)benzoic acid is 3.6. Is the trifluoromethyl substituent an activating or deactivating group in electrophilic aromatic substitution?

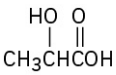
Strategy

Decide whether *p*-(trifluoromethyl)benzoic acid is stronger or weaker than benzoic acid. A substituent that strengthens the acid is a deactivating group because it withdraws electrons, and a substituent that weakens the acid is an activating group because it donates electrons.

Solution

A pK_a of 3.6 means that *p*-(trifluoromethyl)benzoic acid is stronger than benzoic acid, whose pK_a is 4.19. Thus, the trifluoromethyl substituent favors dissociation by helping stabilize the negative charge. Trifluoromethyl must therefore be an electron-withdrawing, deactivating group.

PROBLEM 20-6 Which would you expect to be a stronger acid, the lactic acid found in tired muscles or acetic acid? Explain.



Lactic acid

PROBLEM 20-7 Dicarboxylic acids have two dissociation constants, one for the initial dissociation into a monoanion and one for the second dissociation into a dianion. For oxalic acid, $HO_2C–CO_2H$, the first ionization constant is pK_{a1} = 1.2 and the second ionization constant is pK_{a2} = 4.2. Why is the second carboxyl group far less acidic than the first?

PROBLEM 20-8 The pK_a of *p*-cyclopropylbenzoic acid is 4.45. Is cyclopropylbenzene likely to be more reactive or less reactive than benzene toward electrophilic bromination? Explain.

PROBLEM 20-9 Rank the following compounds in order of increasing acidity. Don't look at a table of pK_a data to help with your answer.

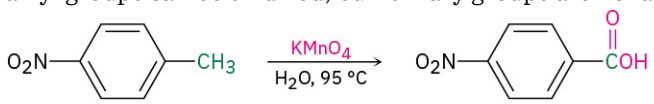
(a) Benzoic acid, *p*-methylbenzoic acid, *p*-chlorobenzoic acid

(b) *p*-Nitrobenzoic acid, acetic acid, benzoic acid

20.5 Preparing Carboxylic Acids

Let's review briefly some of the methods for preparing carboxylic acids that we've seen in previous chapters.

- Oxidation of a substituted alkylbenzene with $KMnO_4$ gives a substituted benzoic acid (**Section 16.8**). Both primary and secondary alkyl groups can be oxidized, but tertiary groups are not affected.

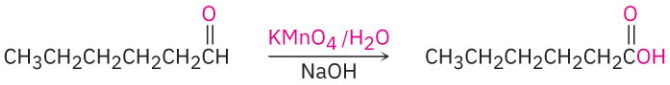


***p*-Nitrotoluene** ***p*-Nitrobenzoic acid (88%)**

- Oxidation of a primary alcohol or an aldehyde yields a carboxylic acid (**Section 17.7** and **Section 19.3**). Primary alcohols are often oxidized with the Dess Martin periodinane, and aldehydes are similarly oxidized with alkaline $KMnO_4$.



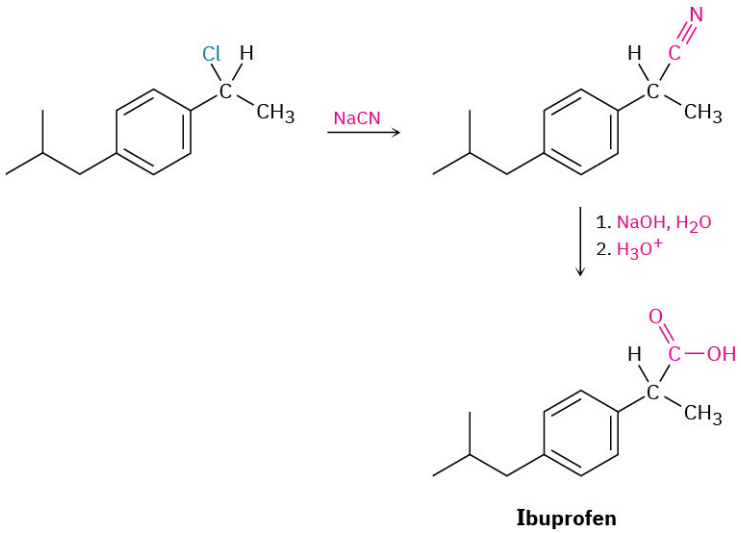
4-Methyl-1-pentanol **4-Methylpentanoic acid**



Hexanal **Hexanoic acid**

Hydrolysis of Nitriles

Carboxylic acids can be prepared from nitriles on heating with aqueous acid or base by a mechanism that we'll discuss in Section 20.7. Since nitriles themselves are usually made by S_N2 reaction of a primary or secondary alkyl halide with CN^-, the two-step sequence of cyanide displacement followed by nitrile hydrolysis is a good way to make a carboxylic acid from an alkyl halide ($RBr \rightarrow RC{\equiv}N \rightarrow RCO_2H$). Note that the product acid has one more carbon than the starting alkyl halide. One example occurs in a commercial route for the synthesis of the nonsteroidal anti-inflammatory drug ibuprofen. (See **Chapter 15 *Chemistry Matters***.)

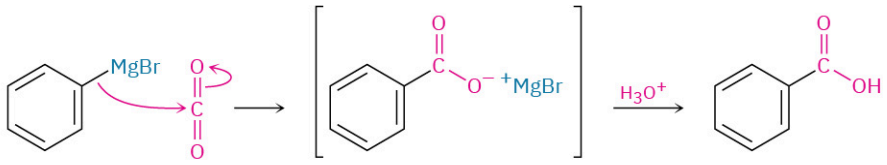


Ibuprofen

Carboxylation of Grignard Reagents

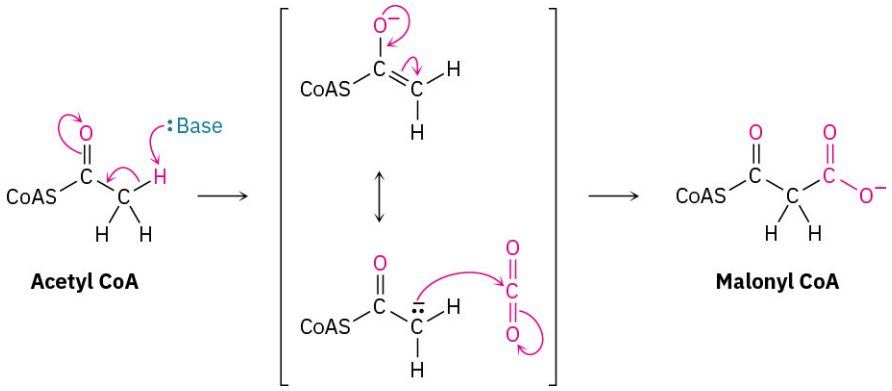
Another method for preparing carboxylic acids is by reaction of a Grignard reagent with CO_2 to yield a metal carboxylate, followed by protonation to give a carboxylic acid. This **carboxylation** reaction is usually carried out by bubbling a stream of dry CO_2 gas through a solution of the Grignard reagent. The organomagnesium halide adds to a $C{=}O$ bond of carbon dioxide in a typical nucleophilic carbonyl addition reaction, and protonation of

the carboxylate by addition of aqueous HCl in a separate step then gives the free carboxylic acid. For example:



Phenylmagnesium bromide **Benzoic acid**

As noted previously, there are no Grignard reagents inside living cells, but there are other types of stabilized carbanions that are often carboxylated. One of the initial steps in fatty-acid biosynthesis, for instance, involves the formation of a carbanion from acetyl CoA, followed by carboxylation to yield malonyl CoA.



Acetyl CoA **Malonyl CoA**

WORKED EXAMPLE 20.2

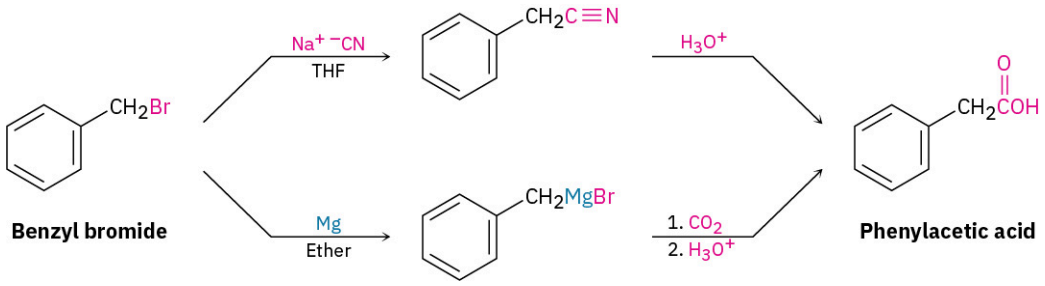
Devising a Synthesis Route for a Carboxylic Acid

How would you prepare phenylacetic acid ($PhCH_2CO_2H$) from benzyl bromide ($PhCH_2Br$)?

Strategy

We've seen two methods for preparing carboxylic acids from alkyl halides: (1) cyanide ion displacement followed by hydrolysis and (2) formation of a Grignard reagent followed by carboxylation. The first method involves an S_N2 reaction and is therefore limited to use with primary and some secondary alkyl halides. The second method involves formation of a Grignard reagent and is therefore limited to use with organic halides that have no acidic hydrogens or reactive functional groups elsewhere in the molecule. In the present instance, either method would work well.

Solution



Benzyl bromide **Phenylacetic acid**

PROBLEM How would you prepare the following carboxylic acids?

20-10 **(a)** $(CH_3)_3CCO_2H$ from $(CH_3)_3CCl$ **(b)** $CH_3CH_2CH_2CO_2H$ from $CH_3CH_2CH_2Br$

20.6 Reactions of Carboxylic Acids: An Overview

We commented earlier in this chapter that carboxylic acids are similar in some respects to both alcohols and

ketones. Like alcohols, carboxylic acids can be deprotonated to give anions, which are good nucleophiles in S_N2 reactions. Like ketones, carboxylic acids undergo addition of nucleophiles to the carbonyl group. However, carboxylic acids also undergo other reactions characteristic of neither alcohols nor ketones. FIGURE 20.3 shows some of the general reactions of carboxylic acids.

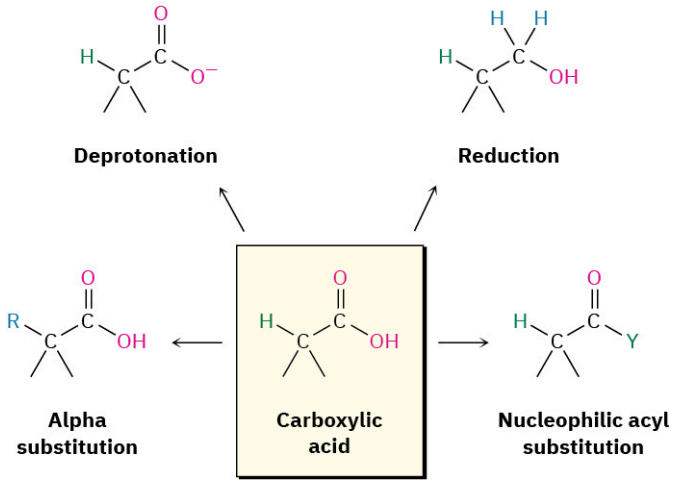
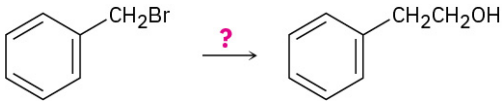


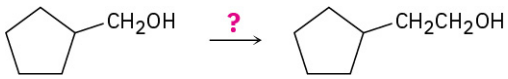
FIGURE 20.3 Some general reactions of carboxylic acids.

Reactions of carboxylic acids can be grouped into the four categories indicated in FIGURE 20.3. Of the four, we've already discussed the acidic behavior of carboxylic acids in Section 20.2 through Section 20.4, and we mentioned reduction by treatment with $LiAlH_4$ in Section 17.4. The remaining two categories are examples of fundamental carbonyl-group reaction mechanisms—nucleophilic acyl substitution and α substitution—that will be discussed in detail in Chapters 21 and 22.

PROBLEM 20-11 How might you prepare 2-phenylethanol from benzyl bromide? More than one step is needed.

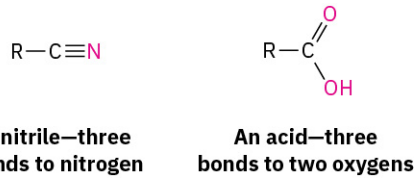


PROBLEM 20-12 How might you carry out the following transformation? More than one step is needed.

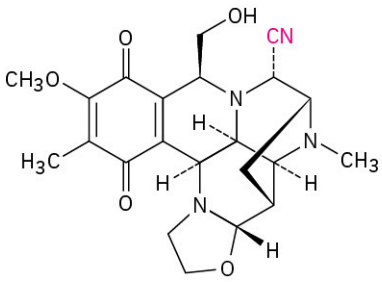


20.7 Chemistry of Nitriles

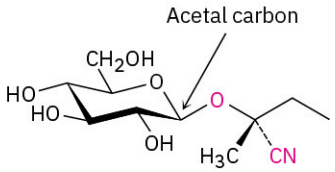
Nitriles are analogous to carboxylic acids in that both have a carbon atom with three bonds to an electronegative atom and both contain a π bond. Thus, some reactions of nitriles and carboxylic acids are similar. Both kinds of compounds are electrophiles, for instance, and both undergo nucleophilic addition reactions.



Nitriles occur infrequently in living organisms, although several hundred examples are known. Cyanocycline A, for instance, has been isolated from the bacterium *Streptomyces lavendulae* and was found to have both antimicrobial and antitumor activity. In addition, more than 1000 compounds called *cyanogenic glycosides* are known. Derived primarily from plants, cyanogenic glycosides contain a sugar with an acetal carbon, one oxygen of which is bonded to a nitrile-bearing carbon (sugar–O–C–CN). On hydrolysis with aqueous acid, the acetal is cleaved (Section 19.10), generating a cyanohydrin (HO–C–CN), which releases hydrogen cyanide. It's thought that the primary function of cyanogenic glycosides is to protect the plant by poisoning any animal foolish enough to eat it. Lotaustralin from the cassava plant is an example.



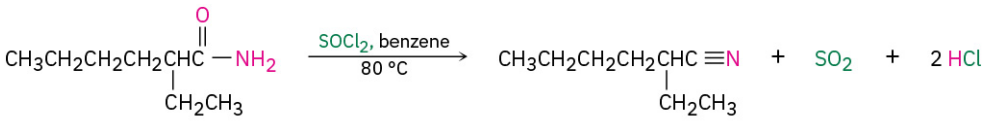
Cyanocycline A



Lotaustralin
(a cyanogenic glycoside)

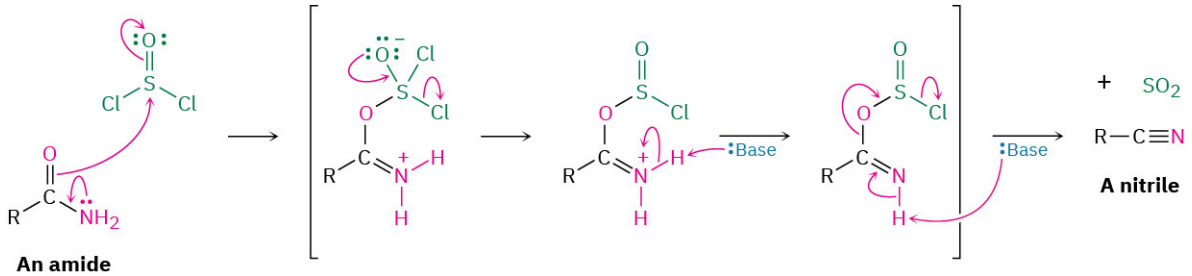
Preparation of Nitriles

The simplest method of nitrile preparation is the S_N2 reaction of CN^- with a primary or secondary alkyl halide, as discussed in **Section 20.5**. Another method for preparing nitriles is by dehydration of a primary amide, $RCONH_2$. Thionyl chloride is often used for this reaction, although other dehydrating agents such as $POCl_3$ also work.



2-Ethylhexanamide **2-Ethylhexanenitrile (94%)**

The dehydration occurs by initial reaction of $SOCl_2$ on the nucleophilic amide oxygen atom, followed by deprotonation and a subsequent E2-like elimination reaction.

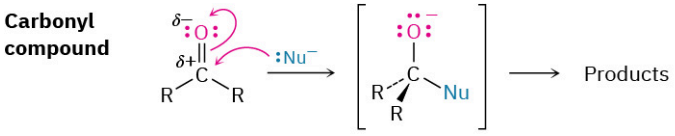


An amide **A nitrile**

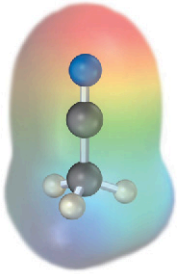
Both methods of nitrile synthesis—S_N2 displacement by CN^- on an alkyl halide and amide dehydration—are useful, but the synthesis from amides is more general because it is not limited by steric hindrance.

Reactions of Nitriles

Like a carbonyl group, a nitrile group is strongly polarized and has an electrophilic carbon atom. Nitriles therefore react with nucleophiles to yield sp^2-hybridized imine anions in a reaction analogous to the formation of a sp^3-hybridized alkoxide ion by nucleophilic addition to a carbonyl group.



Nitrile





Imine anion

Some general reactions of nitriles are shown in **FIGURE 20.4**.

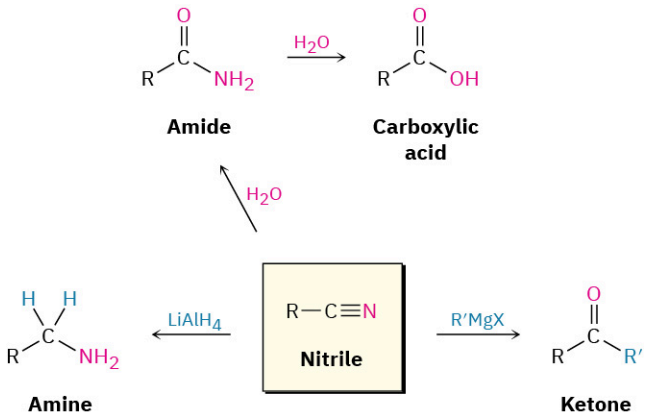
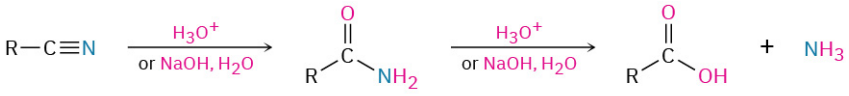


FIGURE 20.4 Some reactions of nitriles.

Hydrolysis: Conversion of Nitriles into Carboxylic Acids

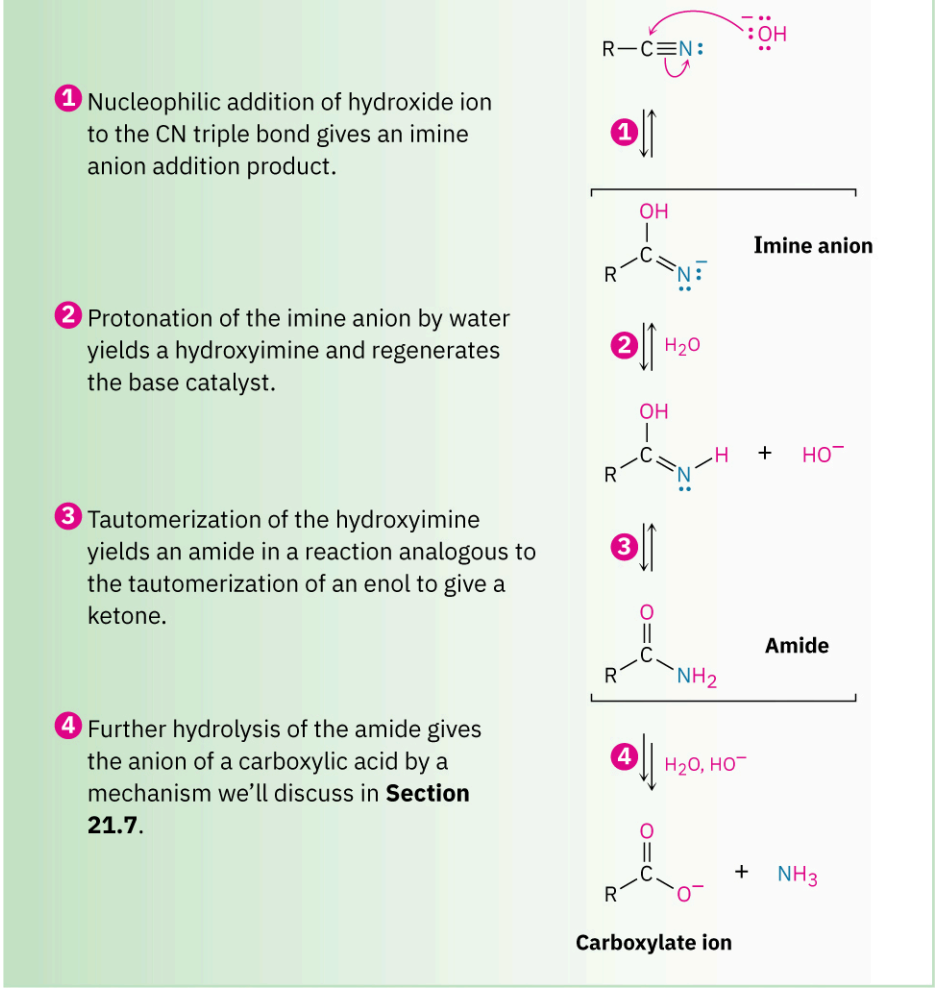
Among the most useful reactions of nitriles is their hydrolysis to yield first an amide and then a carboxylic acid plus ammonia or an amine. The reaction occurs in either basic or acidic aqueous solution:



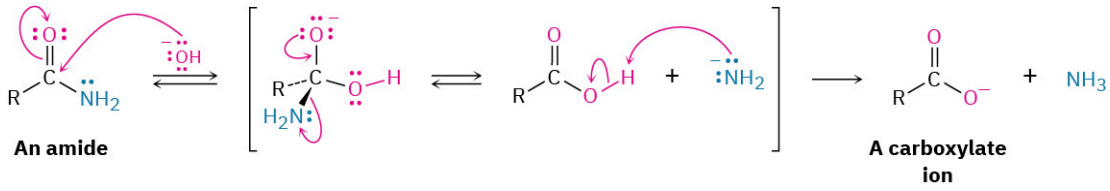
As shown in **FIGURE 20.5**, base-catalyzed nitrile hydrolysis involves nucleophilic addition of hydroxide ion to the polar C≡N bond to give an imine anion in a process similar to the nucleophilic addition to a polar C=O bond to give an alkoxide anion. Protonation then gives a hydroxy imine, which tautomerizes (**Section 9.4**) to an amide in a step similar to the tautomerization of an enol to a ketone. Further hydrolysis gives a carboxylate ion.

FIGURE 20.5 MECHANISM

Mechanism for the basic hydrolysis of a nitrile to yield an amide, which is then hydrolyzed further to a carboxylic acid anion.

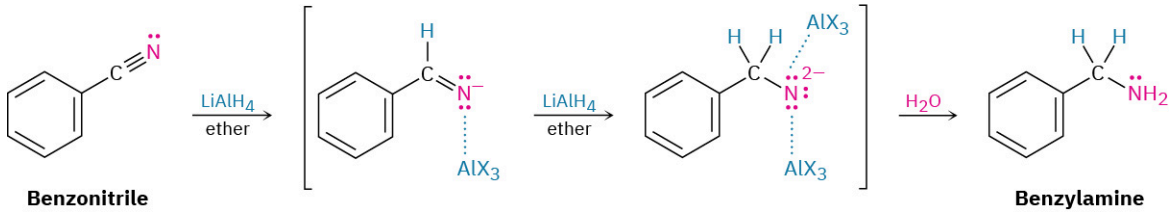


The further hydrolysis of the amide intermediate takes place by a nucleophilic addition of hydroxide ion to the amide carbonyl group, which yields a tetrahedral alkoxide ion. Expulsion of amide ion, NH_2^-, as leaving group gives the carboxylate ion, thereby driving the reaction toward the products. Subsequent acidification in a separate step yields the carboxylic acid. We'll look at this process in more detail in **Section 21.7**.



Reduction: Conversion of Nitriles into Amines

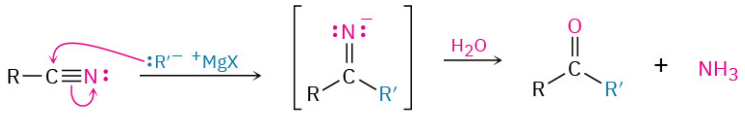
Reduction of a nitrile with $LiAlH_4$ gives a primary amine, RNH_2. The reaction occurs by nucleophilic addition of hydride ion to the polar C≡N bond, yielding an imine anion, which still contains a C=N bond and therefore undergoes a second nucleophilic addition of hydride to give a *dianion*. Both monoanion and dianion intermediates are undoubtedly stabilized by Lewis acid–base complexation to an aluminum species, facilitating the second addition that would otherwise be difficult. Protonation of the dianion by addition of water in a subsequent step gives the amine.



Benzonitrile **Benzylamine**

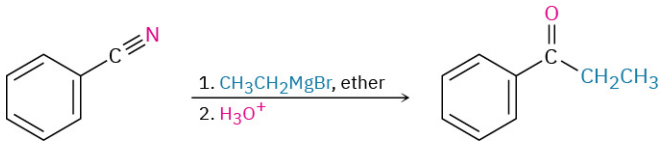
Reaction of Nitriles with Grignard Reagents

Grignard reagents add to a nitrile to give an intermediate imine anion that is hydrolyzed by addition of water to yield a ketone. The mechanism of the hydrolysis is the exact reverse of imine formation (see **FIGURE 19.7**).



Nitrile **Imine anion** **Ketone**

This reaction is similar to the reduction of a nitrile to an amine, except that only one nucleophilic addition occurs rather than two and the attacking nucleophile is a carbanion (R:⁻) rather than a hydride ion. For example:

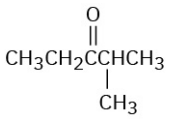


Benzonitrile **Propiophenone (89%)**

WORKED EXAMPLE 20.3

Synthesizing a Ketone from a Nitrile

How would you prepare 2-methyl-3-pentanone from a nitrile?



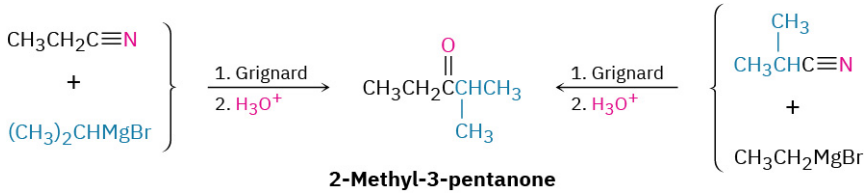
2-Methyl-3-pentanone

Strategy

A ketone results from the reaction between a Grignard reagent and a nitrile, with the C≡N carbon of the nitrile becoming the carbonyl carbon. Identify the two groups attached to the carbonyl carbon atom in the product. One will come from the Grignard reagent and the other will come from the nitrile.

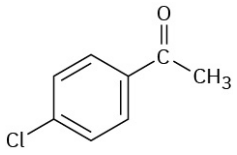
Solution

There are two possibilities.



2-Methyl-3-pentanone

PROBLEM 20-13 How would you prepare the following carbonyl compounds from a nitrile?

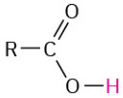
(a) $CH_3CH_2\overset{\overset{O}{\|}}{C}CH_2CH_3$ **(b)** 

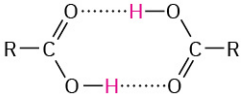
PROBLEM 20-14 How would you prepare 1-phenyl-2-butanone, $C_6H_5CH_2COCH_2CH_3$, from benzyl bromide, $C_6H_5CH_2Br$? More than one step is needed.

20.8 Spectroscopy of Carboxylic Acids and Nitriles

Infrared Spectroscopy

Carboxylic acids have two characteristic IR absorptions that make the $-CO_2H$ group easily identifiable. The O–H bond of the carboxyl group gives rise to a very broad absorption over the range 2500 to 3300 cm^{-1}. The C=O bond shows an absorption between 1710 and 1760 cm^{-1}. The exact position of C=O absorption depends both on the structure of the molecule and on whether the acid is free (monomeric) or hydrogen-bonded (dimeric). Free carboxyl groups absorb at 1760 cm^{-1}, but the more commonly encountered dimeric carboxyl groups absorb in a broad band centered around 1710 cm^{-1}. As with other carbonyl-containing functional groups, conjugation with an alkene or benzene ring lowers the frequency of the C=O stretch by 20 to 30 cm^{-1}.

Free carboxyl (uncommon), 1760 cm^{-1} 

Associated carboxyl (usual case), 1710 cm^{-1} 

Both the broad O–H absorption and the C=O absorption at 1710 cm^{-1} (dimeric) are identified in the IR spectrum of butanoic acid shown in FIGURE 20.6.

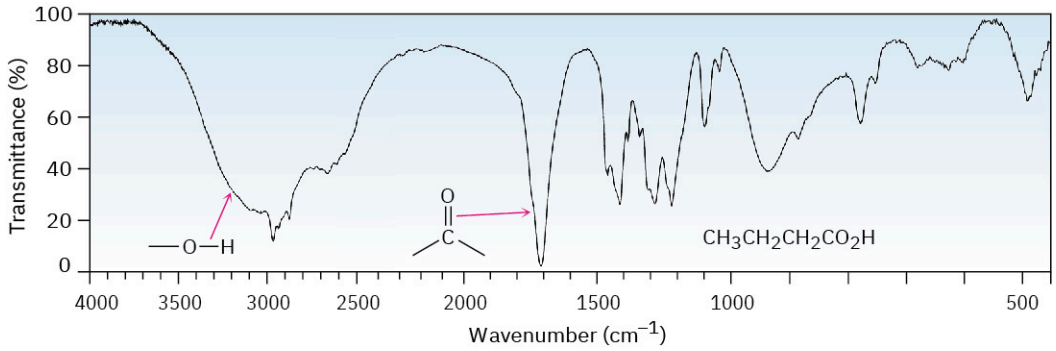


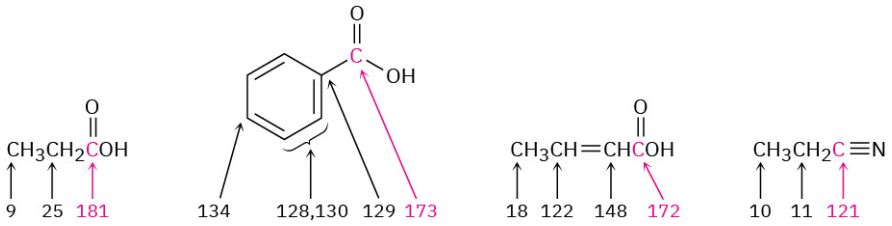
FIGURE 20.6 **IR spectrum of butanoic acid, $CH_3CH_2CH_2CO_2H$.**

Nitriles show an intense and easily recognizable C≡N bond absorption near 2250 cm^{-1} for saturated compounds and 2230 cm^{-1} for aromatic and conjugated molecules. Few other functional groups absorb in this region, so IR spectroscopy is highly diagnostic for nitriles.

PROBLEM 20-15 Cyclopentanecarboxylic acid and 4-hydroxycyclohexanone have the same formula ($C_6H_{10}O_2$), and both contain an –OH and a C=O group. How could you distinguish between them using IR spectroscopy?

Nuclear Magnetic Resonance Spectroscopy

Carboxyl carbon atoms absorb in the range 165 to 185 δ in the ^{13}C NMR spectrum, with aromatic and α,β-unsaturated acids near the upfield end of the range (~165 δ) and saturated aliphatic acids near the downfield end (~185 δ). Nitrile carbons absorb in the range 115 to 130 δ.



In the ^{1}H NMR spectrum, the acidic $-CO_2\mathbf{H}$ proton normally absorbs as a singlet near 12 δ. The chemical shift of the carboxyl proton is concentration and solvent dependent, as these variables can change the extent of hydrogen-bonding in the sample. In some cases, the carboxyl-proton resonance is broadened to the point of being nearly undetectable. Traces of water in the sample can exacerbate the situation. As with alcohols (**Section 17.11**), the $-CO_2H$ proton can be replaced by deuterium when D_2O is added to the sample tube, causing the absorption to disappear from the NMR spectrum. **FIGURE 20.7** shows the ^{1}H NMR spectrum of phenylacetic acid. Note that the $-CO_2H$ absorption occurs at 12.0 δ.

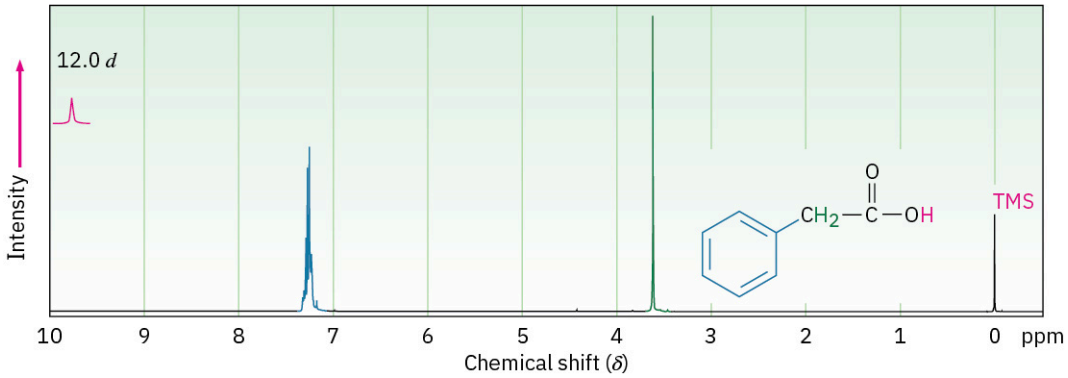


FIGURE 20.7 Proton NMR spectrum of phenylacetic acid, $C_6H_5CH_2CO_2H$.

PROBLEM 20-16 How could you distinguish between the isomers cyclopentanecarboxylic acid and 4-hydroxycyclohexanone by ^{1}H and ^{13}C NMR spectroscopy? (See Problem 20-15.)

CHEMISTRY MATTERS

Vitamin C

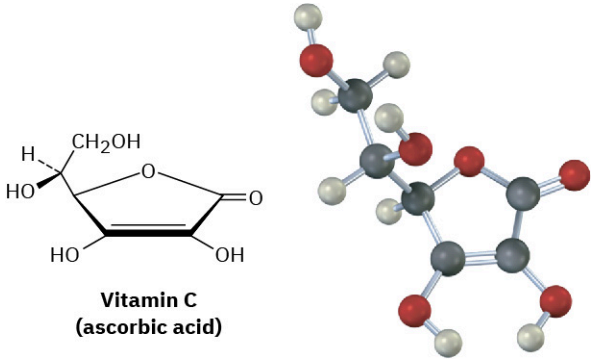
The word *vitamin,* despite its common usage, is actually an imprecise term. Generally speaking, a vitamin is an organic substance that a given organism requires in small amounts to live and grow but is unable to synthesize and must obtain in its diet. Thus, to be considered a vitamin, only a *small* amount of the substance is needed—anywhere from a few micrograms to 100 mg or so per day. Dietary substances needed in larger amounts, such as some amino acids and unsaturated fats, are not considered vitamins.



FIGURE 20.8 In addition to the hazards of weather, participants in early polar expeditions often suffered from scurvy, caused by a dietary vitamin C deficiency. (credit: "Shackleton's expedition to the Antarctic last moments of the Endurance" by Underwood & Underwood/ Wikimedia Commons, Public Domain)

Furthermore, different organisms need different vitamins. More than 4000 species of mammals can synthesize ascorbic acid in their bodies, for instance, but humans are not among them. Ascorbic acid is therefore a human vitamin—what we all know as vitamin C—and must be provided by our diet. Small amounts of more than a dozen other substances are similarly required by humans: retinol (vitamin A), thiamine (vitamin B_1), and tocopherol (vitamin E), for instance.

Vitamin C is surely the best known of all human vitamins. It was the first to be discovered (1912), the first to be isolated (1928), the first to be structurally characterized (1933), and the first to be synthesized in the laboratory (1933). Over 121,000 tons of vitamin C is synthesized worldwide each year, more than the total amount of all other vitamins combined. In addition to its use as a supplement, vitamin C is used as a food preservative, a "flour improver" in bakeries, and an animal-food additive.



Vitamin C (ascorbic acid)

Vitamin C is perhaps most well-known for its antiscorbutic properties, meaning that it prevents the onset of scurvy, a bleeding disease affecting those with a deficiency of fresh vegetables and citrus fruits in their diet. Sailors in the Age of Exploration were particularly susceptible to scurvy, and the death toll was high. The

Portuguese explorer Vasco da Gama lost more than half his crew to scurvy during his two-year voyage around the Cape of Good Hope in 1497–1499.

More recently, large doses of vitamin C have been claimed to prevent the common cold, cure infertility, delay the onset of symptoms in acquired immunodeficiency syndrome (AIDS), and inhibit the development of gastric and cervical cancers. None of these claims have been backed by medical evidence, however. In the largest study yet of the effect of vitamin C on the common cold, a meta-analysis of more than 100 separate trials covering 40,000 people found no difference in the incidence of colds between those who took supplemental vitamin C regularly and those who did not. When taken *during* a cold, however, vitamin C does appear to decrease the cold's duration by perhaps a day.

The industrial preparation of vitamin C involves an unusual blend of biological and laboratory organic chemistry, beginning with glucose and following the five-step route shown in **FIGURE 20.9**. Glucose, a pentahydroxy aldehyde, is first reduced to sorbitol, which is then oxidized by the microorganism *Acetobacter suboxydans*. No chemical reagent is known that is selective enough to oxidize only one of the six alcohol groups in sorbitol, so an enzymatic reaction is used. Treatment with acetone and an acid catalyst then converts four of the other hydroxyl groups into acetal linkages, and the remaining hydroxyl group is chemically oxidized to a carboxylic acid by reaction with aqueous NaOCl (household bleach). Hydrolysis with acid then removes the two acetal groups and causes an internal ester-forming reaction to give ascorbic acid. Each of the five steps has a yield better than 90%.

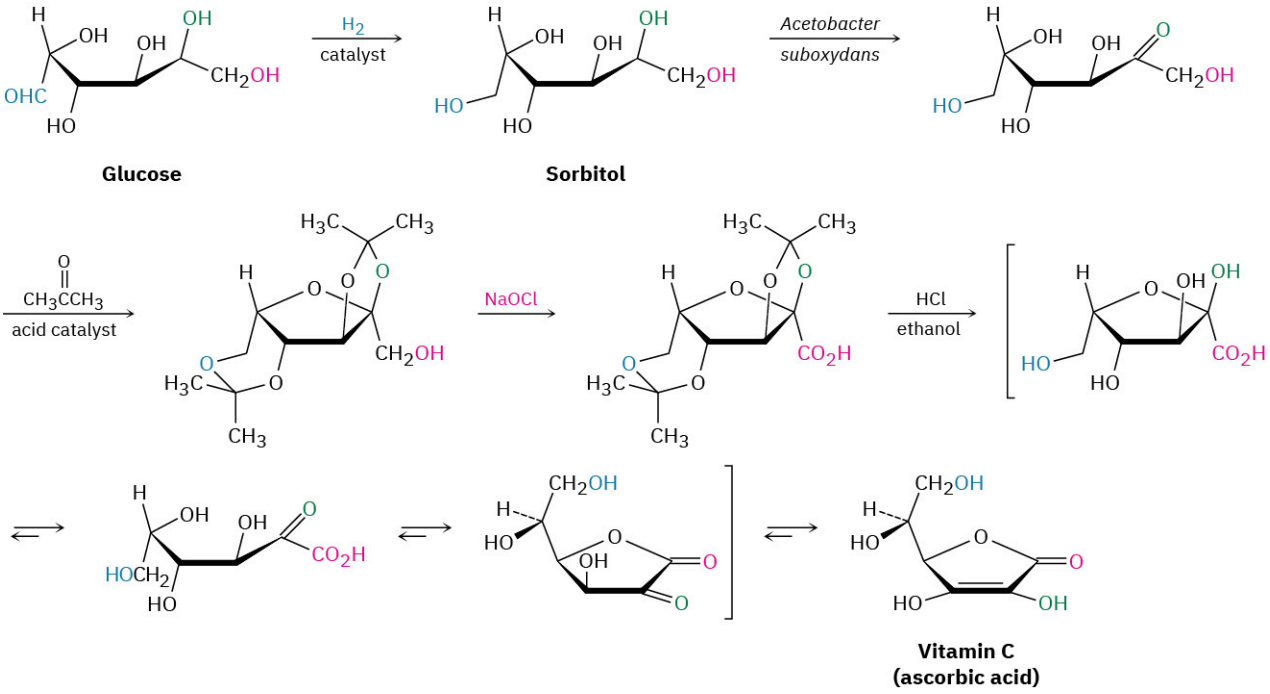


FIGURE 20.9 The industrial synthesis of ascorbic acid from glucose.

Key Terms

- **carboxyl group**
- **carboxylation**
- **carboxylic acid, RCO_2H**
- **Henderson–Hasselbalch equation**
- **nitrile**

Summary

Carboxylic acids are among the most useful building blocks for synthesizing other molecules, both in nature and in the laboratory. Thus, an understanding of their properties and reactions is fundamental to understanding biological chemistry. In this chapter, we've looked both at acids and at their close relatives, **nitriles (RC≡N)**.

Carboxylic acids are named systematically by replacing the terminal *-e* of the corresponding alkane name with *-oic acid*. Like aldehydes and ketones, the carbonyl carbon atom is sp^2-hybridized; like alcohols, carboxylic acids are associated through hydrogen-bonding and therefore have high boiling points.

The distinguishing characteristic of carboxylic acids is their acidity. Although weaker than mineral acids such as HCl, carboxylic acids dissociate much more readily than alcohols because the resultant carboxylate ions are stabilized by resonance between two equivalent forms.

Most carboxylic acids have pK_a values near 5, but the exact pK_a of a given acid depends on structure. Carboxylic acids substituted by electron-withdrawing groups are more acidic (have a lower pK_a) because their carboxylate ions are stabilized. Carboxylic acids substituted by electron-donating groups are less acidic (have a higher pK_a) because their carboxylate ions are destabilized. The extent of dissociation of a carboxylic acid in a buffered solution of a given pH can be calculated with the Henderson–Hasselbalch equation. Inside living cells, where the physiological pH = 7.3, carboxylic acids are entirely dissociated and exist as their carboxylate anions.

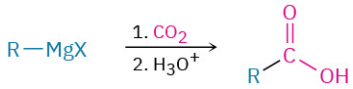
Methods of synthesis for carboxylic acids include (**1**) oxidation of alkylbenzenes, (**2**) oxidation of primary alcohols or aldehydes, (**3**) reaction of Grignard reagents with CO_2 **(carboxylation)**, and (**4**) hydrolysis of nitriles. General reactions of carboxylic acids include (**1**) loss of the acidic proton, (**2**) nucleophilic acyl substitution at the carbonyl group, (**3**) substitution on the α carbon, and (**4**) reduction.

Nitriles are similar in some respects to carboxylic acids and are prepared either by S_N2 reaction of an alkyl halide with cyanide ion or by dehydration of an amide. Nitriles undergo nucleophilic addition to the polar C≡N bond in the same way that carbonyl compounds do. The most important reactions of nitriles are their hydrolysis to carboxylic acids, reduction to primary amines, and reaction with Grignard reagents to yield ketones.

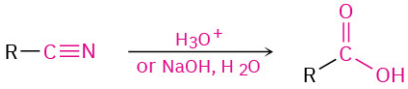
Carboxylic acids and nitriles are easily distinguished spectroscopically. Acids show a characteristic IR absorption at 2500 to 3300 cm^{-1} due to the O–H bond and another at 1710 to 1760 cm^{-1} due to the C=O bond; nitriles have an absorption at 2250 cm^{-1}. Acids also show ^{13}C NMR absorptions at 165 to 185 δ and ^{1}H NMR absorptions near 12 δ. Nitriles have a ^{13}C NMR absorption in the range 115 to 130 δ.

Summary of Reactions

1. Preparation of carboxylic acids (**Section 20.5**)
   a. Carboxylation of Grignard reagents



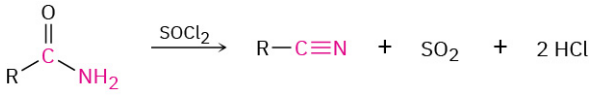
   b. Hydrolysis of nitriles



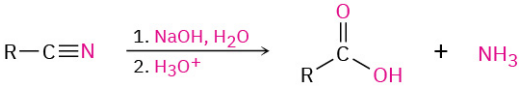
2. Preparation of nitriles (**Section 20.7**)
   a. S_N2 reaction of alkyl halides



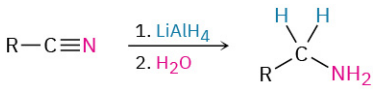
   b. Dehydration of amides



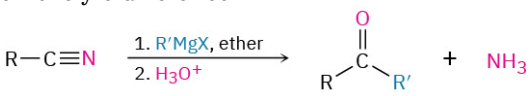
3. Reactions of nitriles (**Section 20.7**)
   a. Hydrolysis to yield carboxylic acids



   b. Reduction to yield primary amines



c. Reaction with Grignard reagents to yield ketones

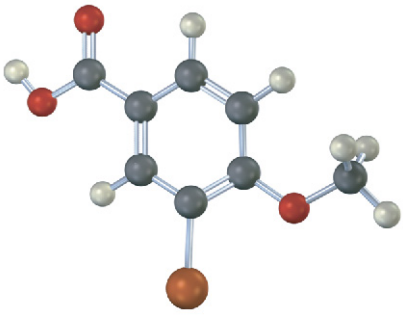


Additional Problems

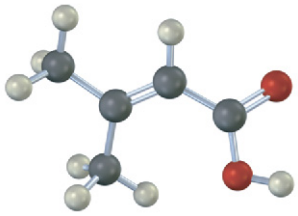
Visualizing Chemistry

PROBLEM 20-17 Give IUPAC names for the following carboxylic acids (reddish brown = Br):

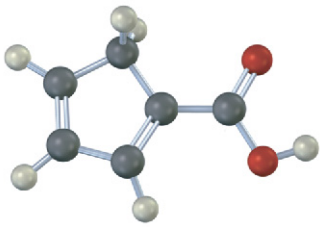
(a)



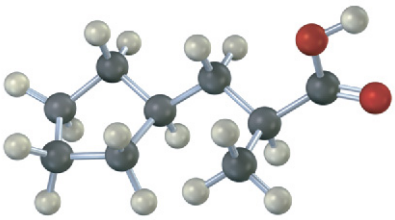
(b)



(c)

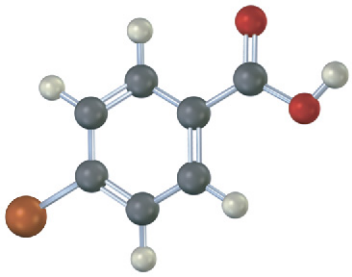


(d)

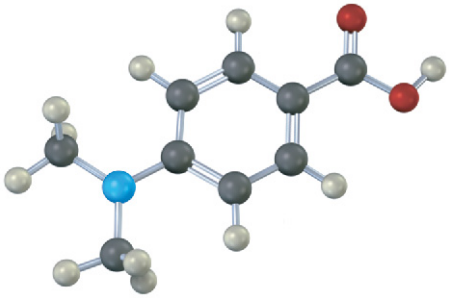


PROBLEM 20-18 Would you expect the following carboxylic acids to be more acidic or less acidic than benzoic acid? Explain. (Reddish brown = Br.)

(a)



(b)



PROBLEM 20-19 The following carboxylic acid can't be prepared from an alkyl halide by either the nitrile hydrolysis route or the Grignard carboxylation route. Explain.

